Lecture #10: Geometry of Linear Programming

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Matrix form of Linear Programming

Last time we ended with the matrix version of the linear programming problem:
Find the vector$x \in \mathbb{R}^n$ that will

A Matrix Form Example

Example

Find the vector $\mathbf{x}$ in $\mathbb{R}^2$ that will

$$\text{maximize } z = \begin{bmatrix} 120 & 100 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \text{ subject to}$$

$$\left.\begin{aligned} \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &\le \begin{bmatrix} 8 \\ 15 \end{bmatrix} \\ \begin{bmatrix} x \\ y \end{bmatrix} &\ge \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}\right\}$$

Constraints and Solutions

Definition

A vector $\mathbf{x}$ in $\mathbb{R}^n$ satisfying the constrains of linear programming problem is called a . A feasible solution that optimizes (either maximizes or minimizes) the objective function of a linear programming problem is called

Finding Feasible Solutions

Consider the linear programming problem above, where the constraints are

$$\begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq \begin{bmatrix} 8 \\ 15 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The vectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

are all feasible solutions because

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Finding Feasible Solutions

Similarly, we can see that

$$\mathbf{x}_4 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \qquad \mathbf{x}_5 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

are not feasible solutions because they violate the linear constraint, i.e.

and

$$\mathbf{x}_6 = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

Geometry of the constraints

Let's look at a single constraint (let's say the i^{th} one) of a linear programming problem in standard form:

can be written as

Definition

The set of points $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ in $\mathbb{R}^n$ that satisfy this constraint is called a . If the inequality is reversed, the set of points $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ in $\mathbb{R}^n$ satisfying

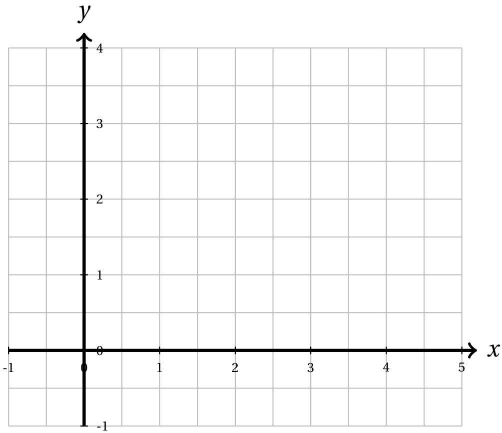
$$\mathbf{a}^\top \mathbf{x} \geq b_i$$

is also called a closed half-space.

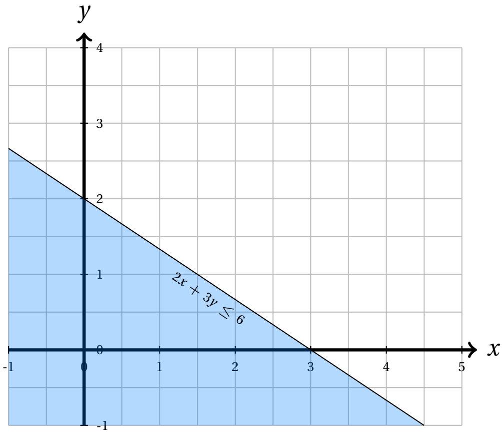
Half Spaces in 2D

Consider the constaint $2x + 3y \leq 6$ and the closed half-space

$$H = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \;\middle|\; \begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \leq 6 \right\}.$$



Answer

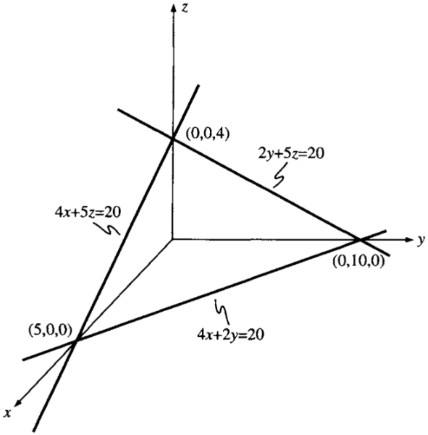


Half Space in 3D

Consider the constaint in 3 dimensions $4x + 2y + 5z \leq 20$ and the closed half-space

$$H = \left\{ \begin{bmatrix} x \\ y \\ y \end{bmatrix} \middle| \begin{bmatrix} 4 & 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \leq 20 \right\}.$$

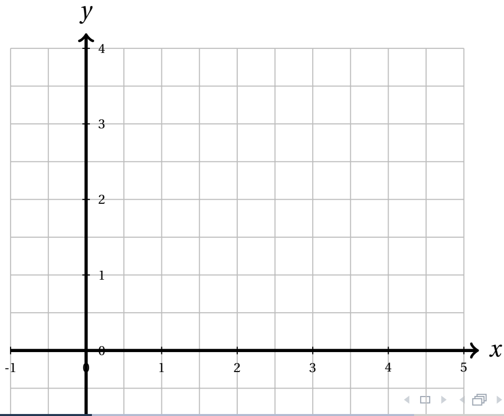
Answer


z
(0,0,4)
2y+5z=20
4x+5z=20
y
(0,10,0)
(5,0,0)
4x+2y=20
x

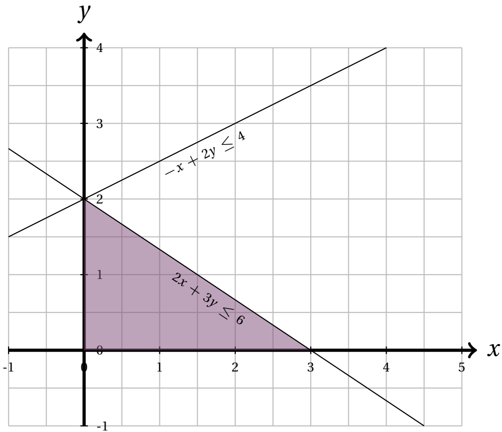
Combining Constraints

Sketch the set of all feasible solutions satisfying the set of inequalities:

$$\begin{aligned} 2x + 3y &\leq 6 \\ -x + 2y &\leq 4 \\ x, y &\geq 0 \end{aligned}$$



Answer



Combining constraints in 3D

Sketch the set of all feasible solutions satisfying the set of inequalities:

$$
\begin{aligned}
5x + 3y + 5z &\leq 15 \\
10x + 4y + 5x &\leq 20 \\
x, y, z &\geq 0
\end{aligned}
$$